

Introduction to the Lab

Students will explore pfsense firewall and Snort Intrusion Detection System (IDS) in this lab. Snort IDS, a signature-based intrusion detection system used to detect network threats. Snort can also function as a simple packet logger. The students will utilize snort as a packet sniffer and develop their own IDS rules for this Lab.

Firewall is device or application that analyzes packet headers and enforces policy based on protocol type, source address, destination address, source port, and destination port. Packets that do not match policy are rejected.



Requirements

- VMware
- Ubuntu Linux (you can use any of your old ubuntu vm running)
- Snort
- Pfsense

Lab breakdown

Part 1: Install and configure snort

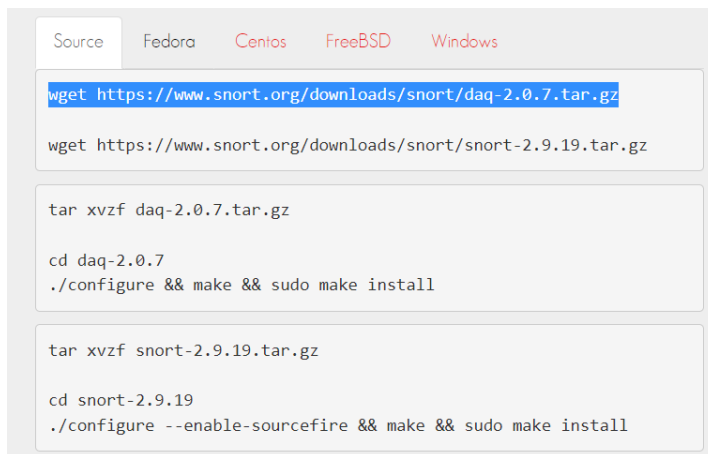
Part 2: Install pfsense and install snort on pfsense

Part 3: Answer questions

Part 1

Install snort on an ubuntu machine. Document your installation including showing a successful running SNORT install.

You can find more information on <https://www.snort.org/#get-started>



Alternative way of installing Snort (easier way/preferred way)

1. Installing snort from source can be very difficult. Run these two commands to make your life easy:

```
sudo apt-get -y update
```

```
sudo apt -y install snort
```

(Attach at least 2 screenshots)

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

```
melvinviado@melvinviado-VMware-Virtual-Platform:~$ sudo apt-get -y update
Hit:1 http://ca.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ca.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:4 http://ca.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
```

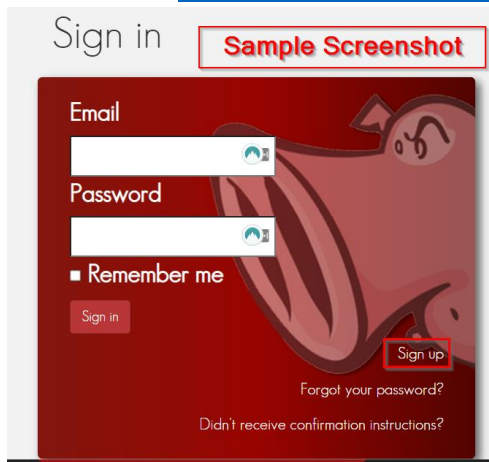
```
melvinviado@melvinviado-VMware-Virtual-Platform:~$ sudo apt -y install snort
[sudo] password for melvinviado:
Files Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libdaq2t64 libdumbnet1 liblua5.1-2 liblua5.1-common
  libnetfilter-queue1 libpcrc3 oinkmaster snort-common snort-common-libraries
  snort-rules-default
Suggested packages:
  snort-doc
```

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

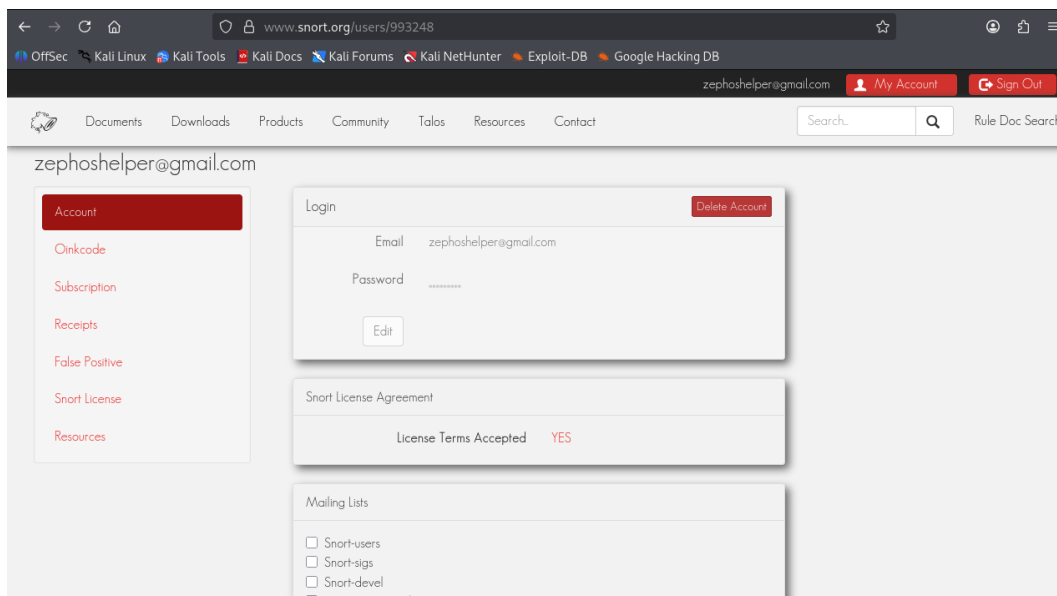
INFT 1202

Total Marks :20

2. Set up a snort account on snort website, this would make it easy to download snort rules. Link: https://www.snort.org/users/sign_in



Attach your screenshot



Set up snort

After installation open configuration files

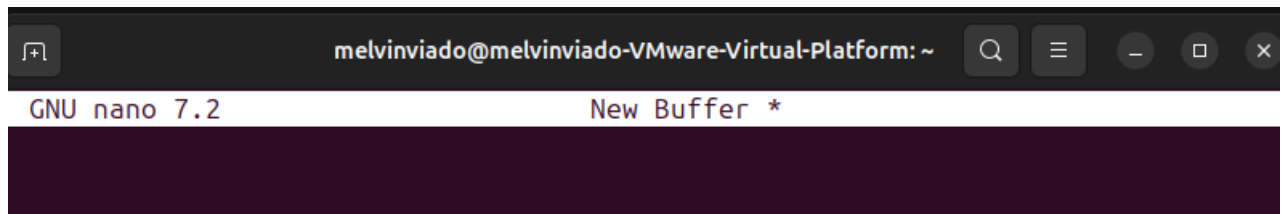
1. The configuration file of snort is stored at `/etc/snort/snort.conf`

Use nano, gedit or vim to edit the file.

(Note, switch to root user and you need to install vim as its not installed by default), Editing this configuration file is optional, but open it and take your screenshot.

Command: `nano /etc/snort/snort.conf`

Attach your screenshot



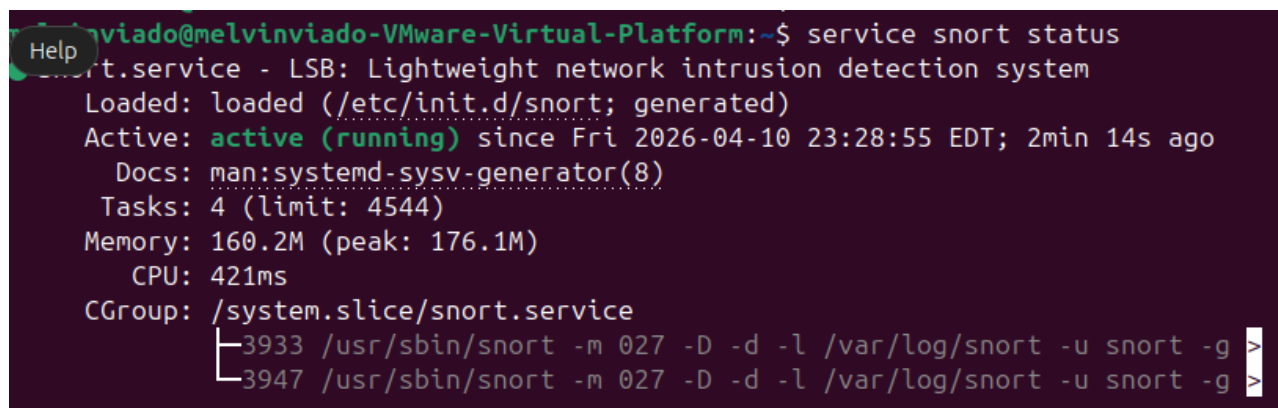
2. After installing the Snort, check if snort is running. run the command to check if snort service is running `service snort status`. If service is not running, use the below command to start service

`$ service snort start`

or

`$ /etc/init.d/snort start`

Attach your screenshot



Snort Rules

Snort is a signature-based IDS, and it defines rules to detect the intrusions. All rules of Snort are stored under `/etc/snort/rules` directory.

1. Run the command `ls /etc/snort/rules`

Attach your screenshot

```
melvinviado@melvinviado-VMware-Virtual-Platform:~$ ls /etc/snort/rules
attack-responses.rules      community-nntp.rules        deleted.rules               netbios.rules              sql.rules
backdoor.rules              community-oracle.rules      dns.rules                   nntp.rules                 telnet.rules
bad-traffic.rules           community-policy.rules      dos.rules                   oracle.rules                tftp.rules
chat.rules                  community-sip.rules          experimental.rules          other-ids.rules            virus.rules
community-bot.rules         community-smtp.rules        exploit.rules                p2p.rules                  web-attacks.rules
community-deleted.rules     community-sql-injection.rules finger.rules                 policy.rules                web-cgi.rules
community-dos.rules         community-virus.rules       ftp.rules                    pop2.rules                  web-client.rules
community-exploit.rules     community-web-attacks.rules icmp-info.rules             pop3.rules                  web-coldfusion.rules
community-ftp.rules         community-web-cgi.rules     icmp.rules                   porn.rules                  web-frontpage.rules
community-game.rules        community-web-client.rules  imap.rules                   rpc.rules                   web-iis.rules
community-icmp.rules        community-web-dos.rules     info.rules                   rservices.rules            web-misc.rules
community-imap.rules        community-web-iis.rules     local.rules                  scan.rules                  web-php.rules
community-inappropriate.rules community-web-misc.rules    misc.rules                   shellcode.rules            x11.rules
community-mail-client.rules community-web-php.rules     multimedia.rules             smtp.rules
community-misc.rules        ddos.rules                  mysql.rules                   snmp.rules
```

2. Run the command `apt install vim` to install vim on your vm
3. Then run the command `vim /etc/snort/rules/web-misc.rules`. This would open one of the rules, you can open any rule of your choice.

Attach your screenshot

```
# Copyright 2001-2005 Sourcefire, Inc. All Rights Reserved
#
# This file may contain proprietary rules that were created, tested and
# certified by Sourcefire, Inc. (the "VRT Certified Rules") as well as
# rules that were created by Sourcefire and other third parties and
# distributed under the GNU General Public License (the "GPL Rules"). The
# VRT Certified Rules contained in this file are the property of
# Sourcefire, Inc. Copyright 2005 Sourcefire, Inc. All Rights Reserved.
# The GPL Rules created by Sourcefire, Inc. are the property of
# Sourcefire, Inc. Copyright 2002-2005 Sourcefire, Inc. All Rights
# Reserved. All other GPL Rules are owned and copyrighted by their
# respective owners (please see www.snort.org/contributors for a list of
# owners and their respective copyrights). In order to determine what
# rules are VRT Certified Rules or GPL Rules, please refer to the VRT
# Certified Rules License Agreement.
#
#
# $Id: web-misc.rules,v 1.118.2.8.2.6 2005/07/22 19:19:54 mwatchinski Exp $
#-----
# WEB-MISC RULES
#-----

alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS 443 (msg:"WEB-MISC SSLv2 Client_Hello with pad Challenge Length overflow at tempt"; flow:to_server,established; flowbits:isnotset,ssl2.client_hello.request; flowbits:isnotset,ssl3.client_hello.request; flowbits:isnotset,tls1.client_hello.request; byte_test:1,<,128,0; content:"|01|"; depth:1; offset:3; byte_test:2,<,768,4; flowbits:set,ssl2.client_hello.request; byte_test:2,>,32,10; classtype:attempted-admin; sid:2657; rev:8;)
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS 443 (msg:"WEB-MISC SSLv2 Client_Hello Challenge Length overflow attempt"; flow:to_server,established; flowbits:isnotset,ssl2.client_hello.request; flowbits:isnotset,ssl3.client_hello.request; flowbits:isnotset,tls1.client_hello.request; byte_test:1,>,127,0; content:"|01|"; depth:1; offset:2; byte_test:2,<,768,3; flowbits:set,ssl2.client_hello.request; byte_test:2,>,32,9; classtype:attempted-admin; sid:2656; rev:7;)
@@@
```

Writing and Adding a Snort Rule

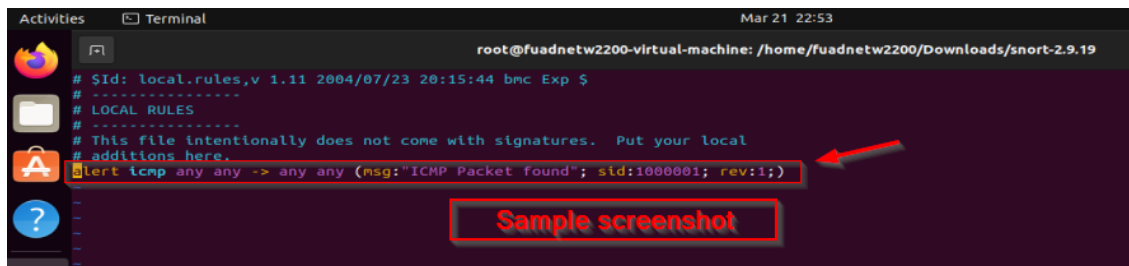
You should add your own rules at `/etc/snort/rules/local.rules`.

Run the command `vim /etc/snort/rules/local.rules` (you can use also use nano or gedit)

Note: for vim, type the **i** key to insert on vim and press the escape key and type the command **:wq** to quit and save

Add the following line into the local.rules file

`alert icmp any any -> any any (msg:"ICMP Packet found"; sid:1000001; rev:1;)`



Attach your screenshot

```
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert icmp any any -> any any (msg:"ICMP Packet found"; sid:1000001; rev:1;)
```

To make the rule become effective, you need to restart the snort service by typing the following command.

`$ sudo service snort restart`

or

`$ sudo /etc/init.d/snort restart`

Triggering an Alert for the New Rule

To trigger an alert for the new rule, you only need to send an ICMP message such as a ping to the VM image where snort is running.

1. First, you need to find the IP address of the VM snort is installed and confirm the IP address of your attacker machine (kali) by typing **Ifconfig**

```
fuad@ubuntu:~/Desktop$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.80.226 netmask 255.255.255.0 broadcast 192.168.80.255
    inet6 fe80::af2e:ee20:88ef:e805 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:3c:8c:f6 txqueuelen 1000 (Ethernet)
    RX packets 3024 bytes 3635280 (3.6 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 975 bytes 101151 (101.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```



```
[03/23/22] fuad@h:\w$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.80.181 netmask 255.255.255.0 broadcast 192.168.80.255
    ether 00:0c:29:07:da:4a txqueuelen 1000 (Ethernet)
    RX packets 626672 bytes 148531034 (141.6 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1148854 bytes 100154401 (95.5 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.44.129 netmask 255.255.255.0 broadcast 192.168.44.255
    inet6 fe80::20c:29ff:fe07:da54 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:07:da:54 txqueuelen 1000 (Ethernet)
    RX packets 3627 bytes 374785 (366.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9632 bytes 937295 (915.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 276793 bytes 24374523 (23.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 276793 bytes 24374523 (23.2 MiB)
```



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

Attach your screenshot of Ubuntu machine IP

```
melvinviado@melvinviado-VMware-Virtual-Platform:~$ ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.74.131 netmask 255.255.255.0 broadcast 192.168.74.255
    inet6 fe80::20c:29ff:fe33:dcb3 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:33:dc:b3 txqueuelen 1000 (Ethernet)
    RX packets 180 bytes 22255 (22.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 145 bytes 15404 (15.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Attach your screenshot of kali machine IP

```
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.74.130 netmask 255.255.255.0 broadcast 192.168.74.255
    inet6 fe80::ee6f:5edd:e606:3bb prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:1d:df:02 txqueuelen 1000 (Ethernet)
    RX packets 752 bytes 67608 (66.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 888 bytes 81240 (79.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

2. Open a terminal in your Kali machine and ping your ubuntu vm running snort or ping your ubuntu from any other vm in your lab

```
[03/23/22] fuad@h:\w$ ping 192.168.80.226
PING 192.168.80.226 (192.168.80.226) 56(84) bytes of data.
64 bytes from 192.168.80.226: icmp_seq=1 ttl=64 time=0.436 ms
64 bytes from 192.168.80.226: icmp_seq=2 ttl=64 time=1.07 ms
64 bytes from 192.168.80.226: icmp_seq=3 ttl=64 time=2.80 ms
64 bytes from 192.168.80.226: icmp_seq=4 ttl=64 time=0.452 ms
^C
--- 192.168.80.226 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3017ms
rtt min/avg/max/mdev = 0.436/1.187/2.795/0.962 ms

[03/23/22] fuad@h:\w$
```

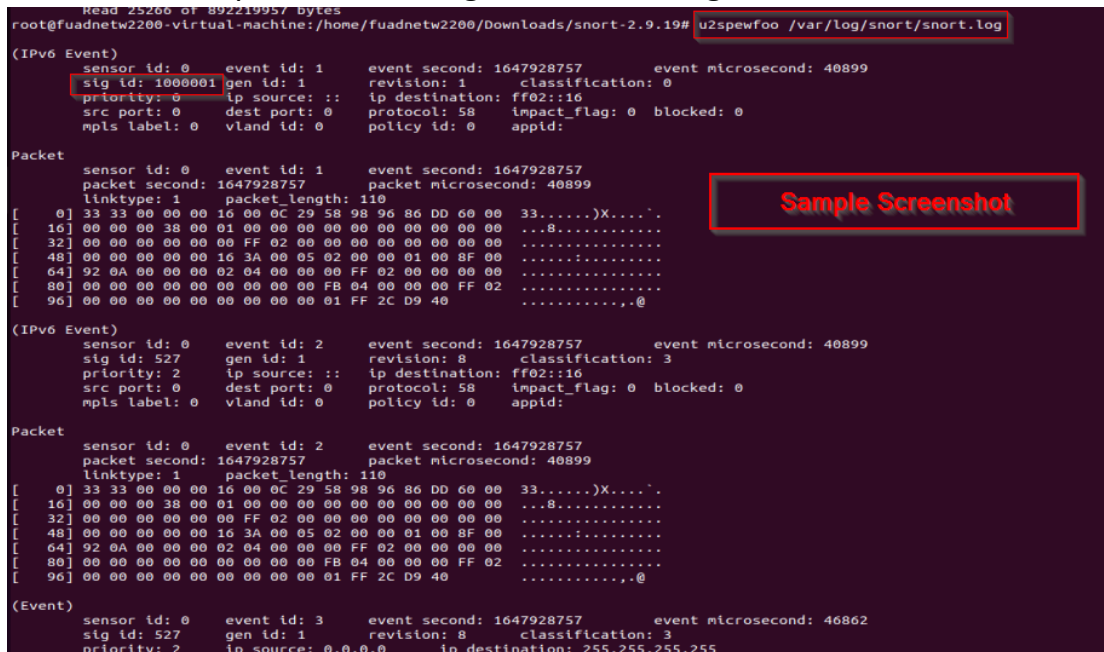
Sample Screenshot

Attach your screenshot

```
(kali@kali)-[~]
$ ping 192.168.74.131
PING 192.168.74.131 (192.168.74.131) 56(84) bytes of data.
64 bytes from 192.168.74.131: icmp_seq=1 ttl=64 time=0.598 ms
64 bytes from 192.168.74.131: icmp_seq=1 ttl=64 time=0.695 ms (DUP!)
64 bytes from 192.168.74.131: icmp_seq=2 ttl=64 time=0.982 ms
64 bytes from 192.168.74.131: icmp_seq=2 ttl=64 time=0.982 ms (DUP!)
64 bytes from 192.168.74.131: icmp_seq=3 ttl=64 time=0.396 ms
64 bytes from 192.168.74.131: icmp_seq=3 ttl=64 time=0.396 ms (DUP!)
64 bytes from 192.168.74.131: icmp_seq=4 ttl=64 time=0.216 ms
64 bytes from 192.168.74.131: icmp_seq=4 ttl=64 time=0.246 ms (DUP!)
64 bytes from 192.168.74.131: icmp_seq=5 ttl=64 time=0.648 ms
64 bytes from 192.168.74.131: icmp_seq=5 ttl=64 time=0.648 ms (DUP!)
64 bytes from 192.168.74.131: icmp_seq=6 ttl=64 time=1.21 ms
64 bytes from 192.168.74.131: icmp_seq=6 ttl=64 time=1.22 ms (DUP!)
64 bytes from 192.168.74.131: icmp_seq=7 ttl=64 time=0.741 ms
64 bytes from 192.168.74.131: icmp_seq=7 ttl=64 time=0.741 ms (DUP!)
64 bytes from 192.168.74.131: icmp_seq=8 ttl=64 time=0.302 ms
64 bytes from 192.168.74.131: icmp_seq=8 ttl=64 time=0.397 ms (DUP!)
64 bytes from 192.168.74.131: icmp_seq=9 ttl=64 time=0.528 ms
64 bytes from 192.168.74.131: icmp_seq=9 ttl=64 time=0.529 ms (DUP!)
^C
--- 192.168.74.131 ping statistics ---
9 packets transmitted, 9 received, +9 duplicates, 0% packet loss, time 8106ms
rtt min/avg/max/mdev = 0.216/0.637/1.215/0.295 ms
```

- After you send the ping messages, the alerts should be triggered, and you can find the log messages in /var/log/snort/snort.log. However, the snort.log file will be binary format. You need to use a tool, called u2spewfoo, to read it. The screenshot below shows the result of reading the snort alerts.

Command: u2spewfoo /var/log/snort/snort.log



```

Read 25266 of 892219957 bytes
root@fuadnetw2200-virtual-machine: /home/fuadnetw2200/Downloads/snort-2.9.19# u2spewfoo /var/log/snort/snort.log

(IPv6 Event)
  sensor id: 0      event id: 1      event second: 1647928757      event microsecond: 40899
  sig id: 1000001  gen id: 1      revision: 1      classification: 0
  priority: 0      ip source: ::      ip destination: ff02::16
  src port: 0      dest port: 0      protocol: 58      impact_flag: 0      blocked: 0
  mpis label: 0    vland id: 0      policy id: 0      appid:

Packet
  sensor id: 0      event id: 1      event second: 1647928757      event microsecond: 40899
  packet second: 1647928757      packet microsecond: 40899
  linktype: 1      packet_length: 110
[ 0] 33 33 00 00 00 16 00 0C 29 58 98 96 86 DD 60 00 33.....)X....
[ 16] 00 00 00 38 00 01 00 00 00 00 00 00 00 00 00 00 ...8.....
[ 32] 00 00 00 00 00 00 FF 02 00 00 00 00 00 00 00 00 .....:.....
[ 48] 00 00 00 00 00 16 3A 00 05 02 00 00 01 00 8F 00 .....:.....
[ 64] 92 0A 00 00 00 02 04 00 00 00 FF 02 00 00 00 00 .....:.....
[ 80] 00 00 00 00 00 00 00 00 00 00 FB 04 00 00 00 FF 02 .....:.....
[ 96] 00 00 00 00 00 00 00 00 01 FF 2C D9 40 .....:..@

(IPv6 Event)
  sensor id: 0      event id: 2      event second: 1647928757      event microsecond: 40899
  sig id: 527      gen id: 1      revision: 8      classification: 3
  priority: 2      ip source: ::      ip destination: ff02::16
  src port: 0      dest port: 0      protocol: 58      impact_flag: 0      blocked: 0
  mpis label: 0    vland id: 0      policy id: 0      appid:

Packet
  sensor id: 0      event id: 2      event second: 1647928757      event microsecond: 40899
  packet second: 1647928757      packet microsecond: 40899
  linktype: 1      packet_length: 110
[ 0] 33 33 00 00 00 16 00 0C 29 58 98 96 86 DD 60 00 33.....)X....
[ 16] 00 00 00 38 00 01 00 00 00 00 00 00 00 00 00 00 ...8.....
[ 32] 00 00 00 00 00 00 FF 02 00 00 00 00 00 00 00 00 .....:.....
[ 48] 00 00 00 00 00 16 3A 00 05 02 00 00 01 00 8F 00 .....:.....
[ 64] 92 0A 00 00 00 02 04 00 00 00 FF 02 00 00 00 00 .....:.....
[ 80] 00 00 00 00 00 00 00 00 00 00 FB 04 00 00 00 FF 02 .....:.....
[ 96] 00 00 00 00 00 00 00 00 01 FF 2C D9 40 .....:..@

(Event)
  sensor id: 0      event id: 3      event second: 1647928757      event microsecond: 46862
  sig id: 527      gen id: 1      revision: 8      classification: 3
  priority: 2      ip source: 0.0.0.0      ip destination: 255.255.255.255
  
```

You can see that the SID is 1000001, and the alerts are generated by the ICMP messages.

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

Attach your screenshot

```
melvinviado@melvinviado-VMware-Virtual-Platform:~$ u2spewfoo /var/log/snort/snort.log

(IPv6 Event)
  sensor id: 0      event id: 1      event second: 1775878694      event microsecond: 177807
  sig id: 1000001  gen id: 1      revision: 1      classification: 0
  priority: 0      ip source: fe80::a441:99f4:30c2:5246      ip destination: ff02::16
  src port: 0      dest port: 0      protocol: 58      impact_flag: 0      blocked: 0
  mpls label: 0    vland id: 0      policy id: 0      appid:

Packet
  sensor id: 0      event id: 1      event second: 1775878694
  packet second: 1775878694      packet microsecond: 177807
  linktype: 1      packet_length: 90
[  0] 33 33 00 00 00 16 00 50 56 C0 00 08 86 DD 60 00 33.....PV.....`.
[ 16] 00 00 00 24 00 01 FE 80 00 00 00 00 00 00 A4 41 ...$......A
[ 32] 99 F4 30 C2 52 46 FF 02 00 00 00 00 00 00 00 00 ..0.RF.....
[ 48] CDROM 00 00 00 00 00 16 3A 00 05 02 00 00 01 00 8F 00 .....:.....
[ 64] AF C8 00 00 00 01 03 00 00 00 FF 02 00 00 00 00 .....
[ 80] 00 00 00 00 00 00 00 01 00 03 .....

(IPv6 Event)
  sensor id: 0      event id: 2      event second: 1775878694      event microsecond: 197339
  sig id: 1000001  gen id: 1      revision: 1      classification: 0
  priority: 0      ip source: fe80::a441:99f4:30c2:5246      ip destination: ff02::16
  src port: 0      dest port: 0      protocol: 58      impact_flag: 0      blocked: 0
  mpls label: 0    vland id: 0      policy id: 0      appid:

Packet
  sensor id: 0      event id: 2      event second: 1775878694
  packet second: 1775878694      packet microsecond: 197339
  linktype: 1      packet_length: 90
[  0] 33 33 00 00 00 16 00 50 56 C0 00 08 86 DD 60 00 33.....PV.....`.
```

4. Next step would be to listen in real time. We would run snort to sniff the traffic in real time. We would run the command below to sniff traffic

Use the command “**snort -A console -c /etc/snort/snort.conf**”

```
root@ubuntu:/etc/snort# snort -A console -c /etc/snort/snort.conf
```

```
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: appid Version 1.1 <Build 5>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPNET Version 1.2 <Build 13>
Commencing packet processing (pid=6349)
03/23-19:25:51.683466 00000001: ICMP Packet found [Priority: 0] {ICMP} 192.168.80.181 -> 192.168.80.226
03/23-19:25:51.683498 00000001: ICMP Packet found [Priority: 0] {ICMP} 192.168.80.226 -> 192.168.80.181
03/23-19:25:52.684175 00000001: ICMP Packet found [Priority: 0] {ICMP} 192.168.80.181 -> 192.168.80.226
03/23-19:25:52.684230 00000001: ICMP Packet found [Priority: 0] {ICMP} 192.168.80.226 -> 192.168.80.181
03/23-19:25:53.699786 00000001: ICMP Packet found [Priority: 0] {ICMP} 192.168.80.181 -> 192.168.80.226
03/23-19:25:53.699849 00000001: ICMP Packet found [Priority: 0] {ICMP} 192.168.80.226 -> 192.168.80.181
03/23-19:25:54.700856 00000001: ICMP Packet found [Priority: 0] {ICMP} 192.168.80.181 -> 192.168.80.226
03/23-19:25:54.700891 00000001: ICMP Packet found [Priority: 0] {ICMP} 192.168.80.226 -> 192.168.80.181
```

Kali IP Address (Attacker Machine) → 192.168.80.181

Ubuntu IP Address of Snort machine → 192.168.80.226

Sample Screenshot

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20

Attach your screenshot

```
melvinviado@melvinviado-VMware-Virtual-Platform: ~  
Preprocessor Object: SF_SSH Version 1.1 <Build 3>  
Preprocessor Object: appid Version 1.1 <Build 5>  
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>  
Preprocessor Object: SF_GTP Version 1.1 <Build 1>  
Preprocessor Object: SF_POP Version 1.0 <Build 1>  
Preprocessor Object: SF_DCEP2 Version 1.0 <Build 3>  
Commencing packet processing (pid=5283)  
4/10-23:39:24.614813  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.130 -> 192.168.74.131  
4/10-23:39:24.614814  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.130 -> 192.168.74.131  
4/10-23:39:24.614844  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:24.614883  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:24.615283  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:24.615284  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:25.618335  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.130 -> 192.168.74.131  
4/10-23:39:25.618338  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.130 -> 192.168.74.131  
4/10-23:39:25.618410  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:25.618489  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:25.619721  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:25.619724  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:26.619922  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.130 -> 192.168.74.131  
4/10-23:39:26.619967  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:26.620067  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.130 -> 192.168.74.131  
4/10-23:39:26.620075  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:26.620379  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:26.620380  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:27.634185  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.130 -> 192.168.74.131  
4/10-23:39:27.634187  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.130 -> 192.168.74.131  
4/10-23:39:27.634224  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:27.634289  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:27.635116  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130  
4/10-23:39:27.635117  [**] [1:1000001:1] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.74.131 -> 192.168.74.130
```


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Total Marks :20

5. Use ctrl + C to end your sniffing and run it again, this time you will run an nmap scan against your target machine. Use the nmap command **nmap -sC -sV [UbuntuIP]**. Once you run this, wait for some time for snort to display the logs.

```
[03/23/22] fuad@h:\w$ nmap -sC -sV 192.168.80.226
Starting Nmap 7.92 ( https://nmap.org ) at 2022-03-23 23:17 EDT
Nmap scan report for 192.168.80.226
Host is up (0.00086s latency).
All 1000 scanned ports on 192.168.80.226 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:3C:8C:F6 (VMware)
```

Sample Screenshot

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .
Nmap done: 1 IP address (1 host up) scanned in 0.97 seconds

```
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: apid Version 1.1 <Build 5>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Commencing packet processing (pid=6548)
03/23-19:51:20.445071 [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2] (TCP) 192.168.80.181:46075 -> 192.168.80.226:705
03/23-19:51:20.472918 [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] (TCP) 192.168.80.181:46075 -> 192.168.80.226:161
03/23-19:52:17.880159 [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [Priority: 3] (UDP) 192.168.80.1:610 36 -> 239.255.255.250:1900
03/23-19:52:17.910531 [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [Priority: 3] (UDP) 192.168.80.1:610 40 -> 239.255.255.250:1900
03/23-19:52:18.883055 [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [Priority: 3] (UDP) 192.168.80.1:610 36 -> 239.255.255.250:1900
03/23-19:52:18.913678 [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [Priority: 3] (UDP) 192.168.80.1:610 40 -> 239.255.255.250:1900
03/23-19:52:19.897194 [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [Priority: 3] (UDP) 192.168.80.1:610 36 -> 239.255.255.250:1900
03/23-19:52:19.928804 [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [Priority: 3] (UDP) 192.168.80.1:610 40 -> 239.255.255.250:1900
03/23-19:52:20.908788 [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [Priority: 3] (UDP) 192.168.80.1:610 36 -> 239.255.255.250:1900
03/23-19:52:20.939718 [**] [1:1917:6] SCAN UPnP service discover attempt [**] [Classification: Detection of a Network Scan] [Priority: 3] (UDP) 192.168.80.1:610 40 -> 239.255.255.250:1900
```

Snort detected a Network scan

Snort detected nmap's scan as an attempted information leak

Sample Screenshot

Kali IP

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Attach your screenshot of nmap scan

```
(kali@kali)-[~]
$ nmap -sC -sV 192.168.74.131
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-10 23:40 EDT
Nmap scan report for 192.168.74.131
Host is up (0.0010s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-server-header: Apache/2.4.58 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
8000/tcp   open  http    Splunkd httpd
| http-title: Site doesn't have a title (text/html; charset=UTF-8).
|_Requested resource was http://192.168.74.131:8000/en-US/account/login?return_to=%2Fen-US%2F
| http-robots.txt: 1 disallowed entry
|_/
|_http-server-header: Splunkd
8089/tcp   open  ssl/http Splunkd httpd
|_http-title: splunkd
| ssl-cert: Subject: commonName=SplunkServerDefaultCert/organizationName=SplunkUser
| Not valid before: 2026-02-20T20:43:30
|_Not valid after: 2029-02-19T20:43:30
| http-robots.txt: 1 disallowed entry
|_/
|_http-server-header: Splunkd
MAC Address: 00:0C:29:33:DC:B3 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 44.23 seconds
```

Attach your screenshot of nmap scan detected

```
04/10-23:40:11.564050  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2]
{TCP} 192.168.74.130:45114 -> 192.168.74.131:161
04/10-23:40:11.573134  [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2]
{TCP} 192.168.74.130:45114 -> 192.168.74.131:161
04/10-23:40:11.591282  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2]
{TCP} 192.168.74.130:45114 -> 192.168.74.131:705
04/10-23:40:11.611828  [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2]
{TCP} 192.168.74.130:45114 -> 192.168.74.131:705
```


Part 2

Install Pfsense and install snort module on pfsense

1. Install pfsense on VMware, take screenshots of installation process. **(Attach at least 10 screenshots)**

Useful links for reference, you can follow any other walk through you prefer

Pfsense installation - <https://techexpert.tips/pfsense/pfsense-server-installation/>

Source:

Downloaded from: [Netgate Official Mirror](#)

Automated release created on 2025-05-23 21:13:41 UTC

▼ Assets 5

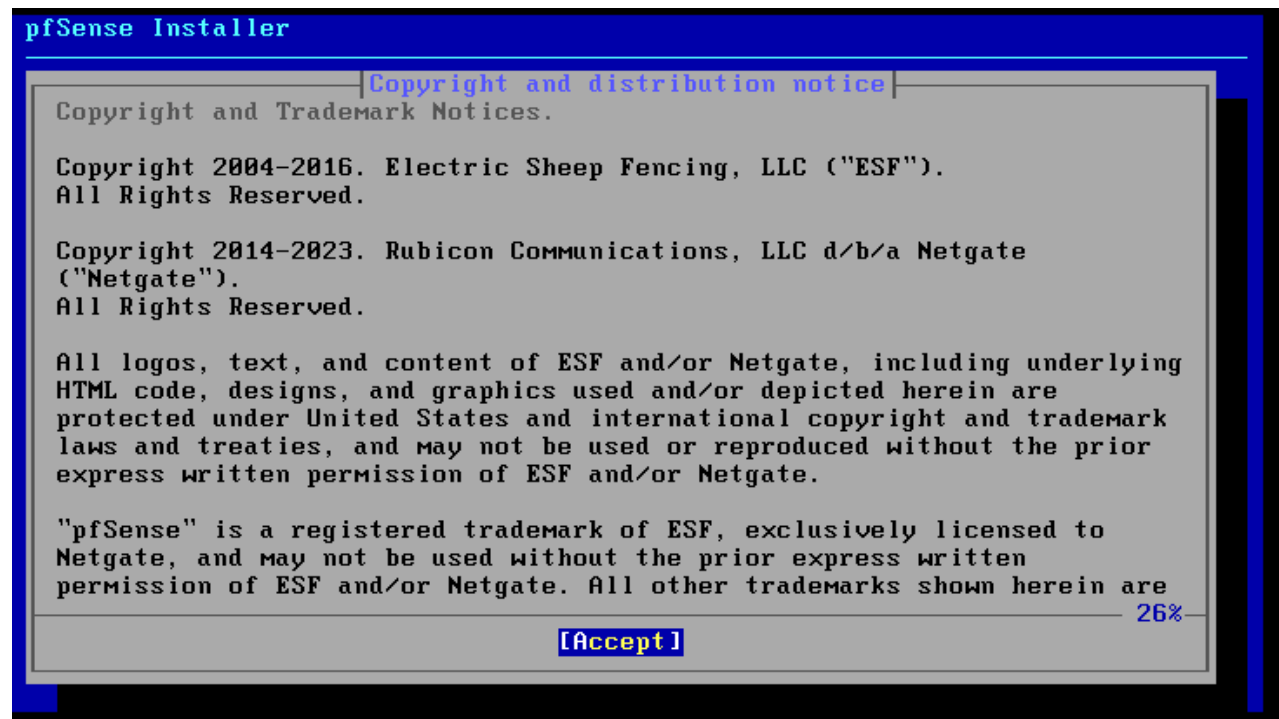
 pfSense-CE-2.7.2-RELEASE-amd64.iso	834 MB	May 23, 2025
--	--------	--------------

 FreeBSD version 10 and earlier 64-bit

Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

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Total Marks :20



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

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Total Marks :20



Virtual Machine Settings

Hardware Options	
Device	Summary
 Memory	2.1 GB
 Processors	1
 Hard Disk (SCSI)	20 GB
 CD/DVD (IDE)	Using file C:\Users\Mel\Downlo...
 Network Adapter	NAT
 Network Adapter 2	Host-only
 USB Controller	Present
 Sound Card	Auto detect
 Display	Auto detect

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```
Starting syslog...done.
Starting CRON... done.
pfSense 2.7.2-RELEASE amd64 20231206-2010
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VMware Virtual Machine - Netgate Device ID: dcd5ae7aaf4b6c92b21f

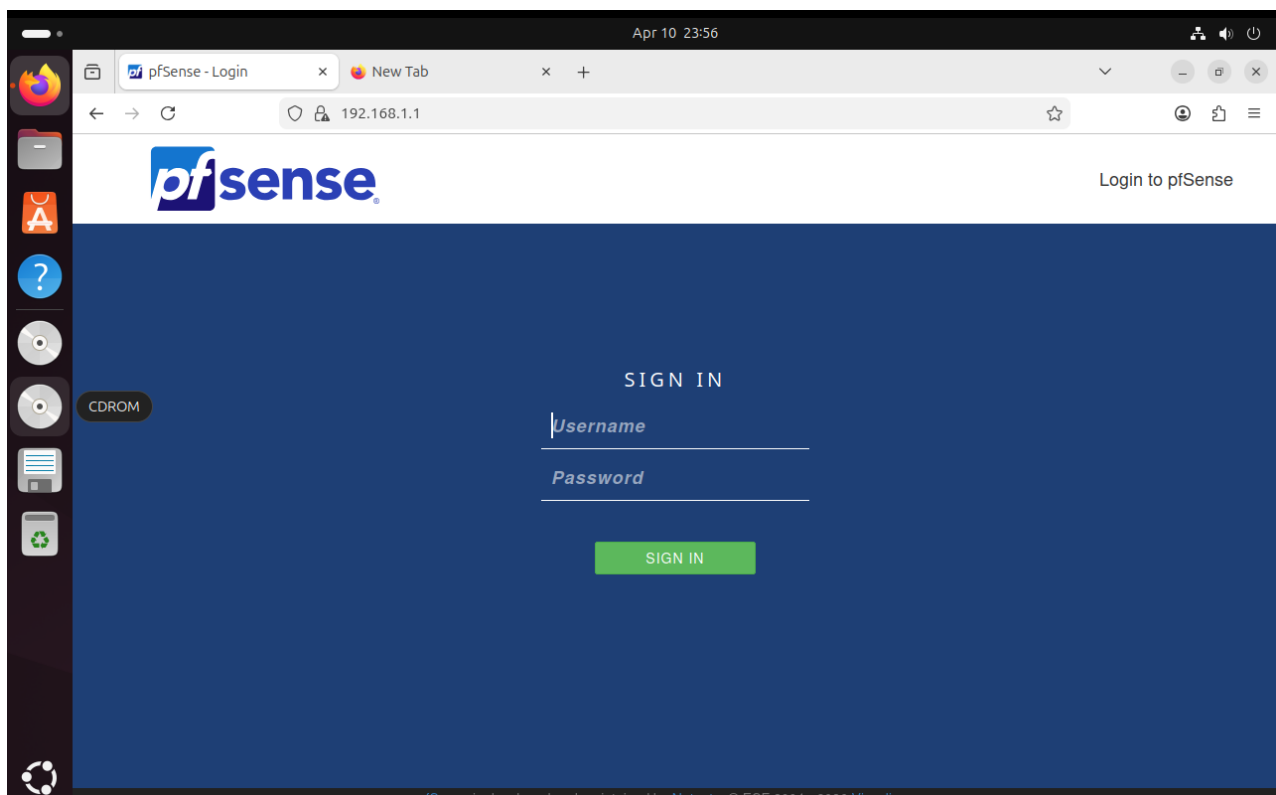
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 172.10.0.139/24
LAN (lan)      -> em1      -> v4: 192.168.1.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: █
```

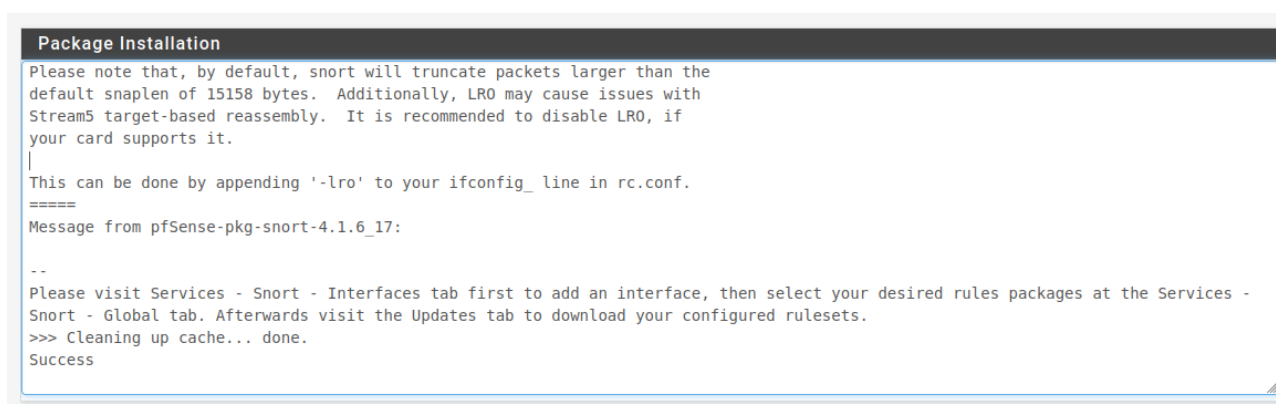
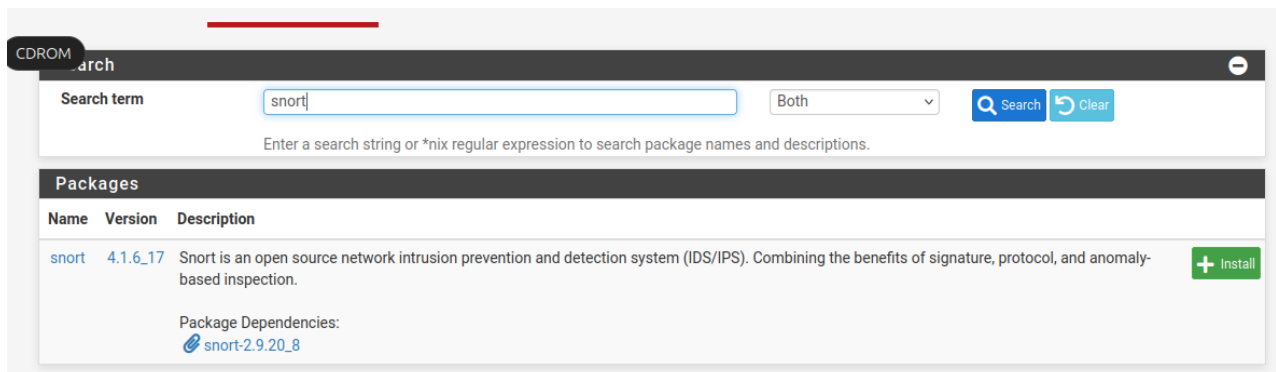
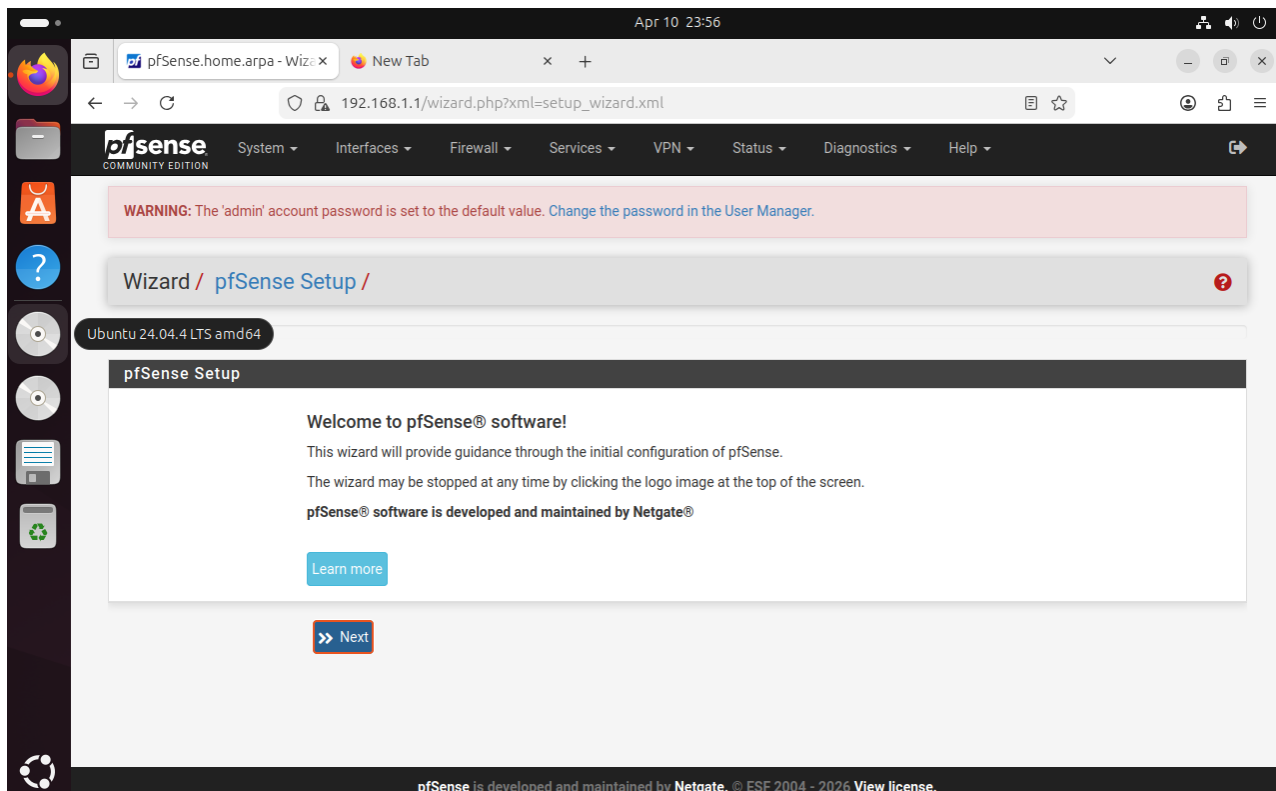
2. Install Snort IDS on your pfsense firewall (**Attach at least 2 screenshots**)



Lab 6: Firewall & Intrusion Detection Systems (Individual Lab)

INFT 1202

Total Marks :20



Part 3

Questions

Answer the questions in the Introduction section and justify your answers.

1. Simple yes or no answer will not get any credits.

- a) What is a zero-day attack?

A zero-day attack are attacks where defenders have zero days to fix the problem because it exploits a vulnerability before the defender knows about it or has a patch ready

- b) Can Snort catch zero-day network attacks? If not, why not? If yes, how?

Snort cannot catch zero-day attacks on its own but it can help detect network traffic that matches patterns of exploits